

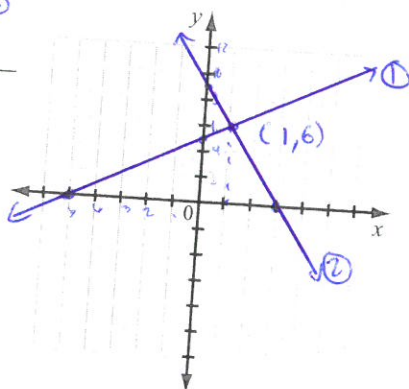
Simultaneous equations

UNIT 3: The graphical method

QUESTION 1 Solve, by drawing graphs, the following pairs of simultaneous equations.

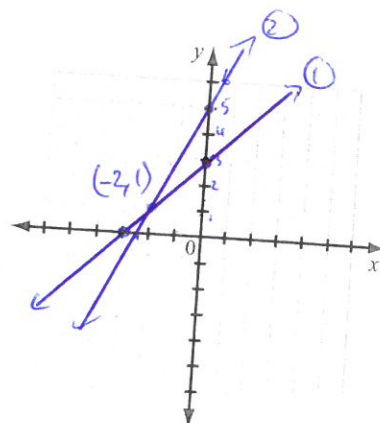
a $y = x + 5$ ①
 $y = -3x + 9$ ②

$$\boxed{\begin{matrix} x = 1 \\ y = 6 \end{matrix}}$$



b $y = x + 3$ ①
 $y = 2x + 5$ ②

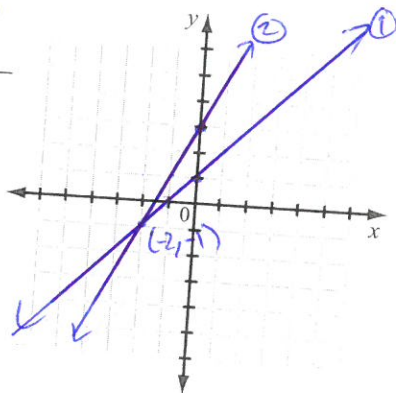
$$\boxed{\begin{matrix} x = -2 \\ y = 1 \end{matrix}}$$



QUESTION 2 Solve graphically the following pairs of simultaneous equations.

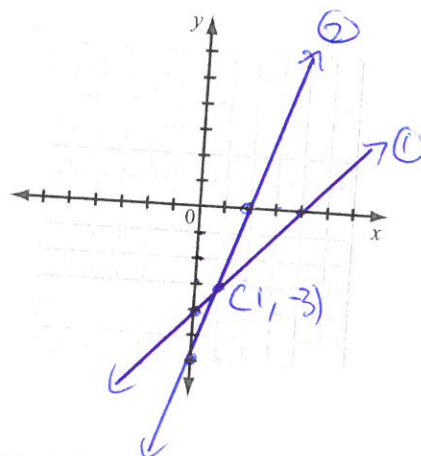
a $y = x + 1$ ①
 $y = 2x + 3$ ②

$$\boxed{\begin{matrix} x = -2 \\ y = -1 \end{matrix}}$$



b $y = x - 4$ ①
 $y = 3x - 6$ ②

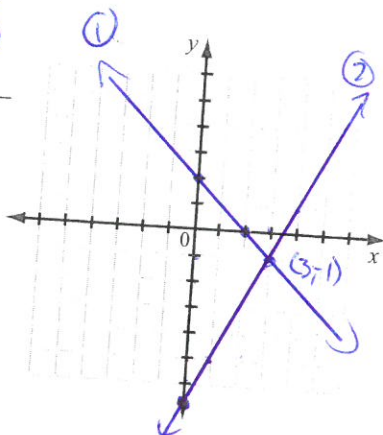
$$\boxed{\begin{matrix} x = 1 \\ y = -3 \end{matrix}}$$



QUESTION 3 Graph each pair of equations on the same number plane to find their solution.

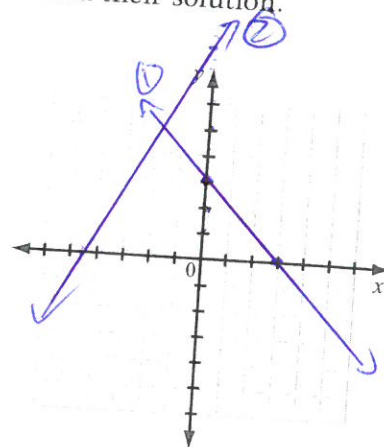
a $y = -x + 2$ ①
 $y = 2x - 7$ ②

$$\boxed{\begin{matrix} x = 3 \\ y = -1 \end{matrix}}$$



b $x + y = 3$ ①
 $2x - y = -9$ ②
 $y = 2x + 9$

$$\boxed{\begin{matrix} x = -2 \\ y = 5 \end{matrix}}$$



Simultaneous equations

UNIT 7: The method of elimination

QUESTION 1 Solve the following pairs of simultaneous equations by the method of elimination.

a $x + 2y = 8$

$+ 3x - 2y = 4$

$4x = 12$

$x = 3$

$x + 2y = 8$

$3 + 2y = 8$

$2y = 5$

$y = 2.5$

$\therefore x = 3, y = 2.5$

b $2x - 3y = 6$

$+ x + 3y = 9$

$3x = 15$

$x = 5$

$5 + 3y = 9$

$3y = 4$

$y = \frac{4}{3}$

$x = 5, y = \frac{4}{3}$

c $2x + 5y = 19$

$+ 3x - 5y = 6$

$5x = 25$

$x = 5$

$15 - 5y = 6$

$-5y = -9$

$y = \frac{9}{5}$

$x = 5, y = \frac{9}{5}$

d $2x + y = 10$

$+ 3x - y = 5$

$5x = 15$

$x = 3$

$6 + y = 10$

$y = 4$

$\therefore x = 3, y = 4$

e $3x + 4y = 14$

$- 3x + 3y = 8 \times 3$

$y = 5$

$3x + 5 = 3$

$x = -2$

$\therefore x = -2, y = 5$

f $2x - 2y = 9 \times 2$

$= x - 2y = 9$

$3x = -3$

$x = -1$

$-1 - 2y = 9$

$-2y = 10$

$y = -5$

$\therefore x = -1, y = -5$

QUESTION 2 Use the method of elimination to solve the following pairs of simultaneous equations.

a $3x + 4y = 12$

$+ 4x - 4y = 16 \times 4$

$7x = 28$

$x = 4$

$4 - y = 4$

$y = 0$

$\therefore x = 4, y = 0$

b $4x + 5y = 22$

$- 10x + 4y = 10 \times 4$

$y = 12$

$4x + 12 = 22$

$x + 12 = 10$

$x = -2$

$\therefore y = 12, x = -2$

c $2x + 3y = 11$

$+ 3x - 3y = -8 \times 3$

$5x = 5$

$x = 1$

$2 + 3y = 11$

$3y = 9$

$y = 3$

$\therefore x = 1, y = 3$

d $5x - 3y = 9$

$+ 9x + 3y = 42 \times 3$

$14x = 21$

$x = \frac{21}{14} = \frac{3}{2} = 1.5$

$5 \times (1.5) - 3y = 9$

$7.5 - 3y = 9$

$-3y = 1.5, y = -0.5$

$\therefore x = 1.5, y = -0.5$

Simultaneous equations

UNIT 8: Mixed questions

QUESTION 1 Solve the following pairs of simultaneous equations by the method of elimination.

a $9a - 2b = 36$

+ $a + 2b = 14$

$10a = 50$

$a = 5$

$5 + 2b = 14$

$2b = 9$

$b = 4.5$

b $2x + 2y = 8 \times 16 \times 2$

- $5x + 2y = -3$

$-x = 19$

$x = -19$

$-38 + y = 8$

$y = 46$

$x = -19$

$y = 46$

c $x + 3y = 8$

+ $9x - 3y = 9 \times 3$

$10x = 35$

$x = 3.5$

$3.5 + 3y = 8$

$3y = 4.5$

$y = 1.5$

$x = 3.5$

$y = 1.5$

QUESTION 2 Use the method of substitution to solve the following pairs of simultaneous equations.

a $2x - y = 9$ ①

$3x - 2y = 15$ ②

① $2x - y = 9 \Rightarrow y = 2x - 9$

①+② $3x - 2(2x - 9) = 15$

$x = 0$

$y = -9$

$x = 0$

$y = -9$

b $3x + 8y = 4$

$3x + 2y = -2 \Rightarrow 3x = -2y - 2$

$-2y - 2 + 8y = 4$

$6y = 6$

$y = 1$

$3x = -2 - 2$

$x = -\frac{4}{3}$

$x = -\frac{4}{3}$

$y = 1$

c $5m - 2n = 20$

$3m - 4n = 12$

$m = 4$

$n = 0$

QUESTION 3 Solve the following simultaneous equations by any suitable method.

a $x + y = 10$

$3x - 4y = 2$

$x = 6$

$y = 4$

b $x + 5y = 15$

$-x + 2y = 6$

$x = 0$

$y = 3$

c $2x + y = -8$

$3x - 2y = -12$

$x = -4$

$y = 0$

d $3x + y = 7$

- $3x + 2y = 9 \times 3$

$-y = -20$

$y = 4$

$3x + 4 = 9$

$x = 1$

$x = 1, y = 4$

e $3x + 5y = 25$

+ $10x - 5y = 840 \times 5$

$13x = 65$

$x = 5$

$15 + 5y = 25$

$5y = 10$

$y = 2$

$y = 2$

$x = 5$

f $2x - 5y = 13$

$5x - 3y = -15$

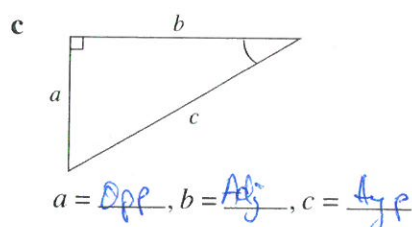
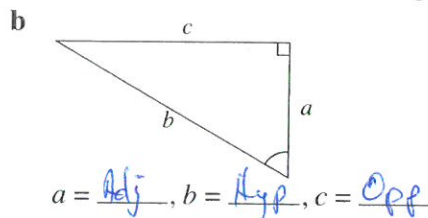
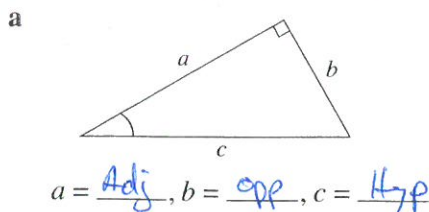
$x = -6$

$y = -5$

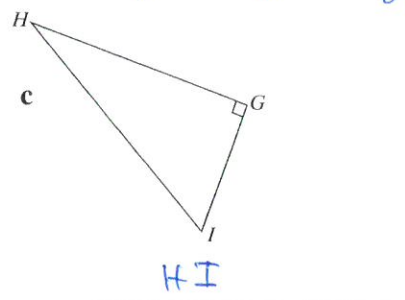
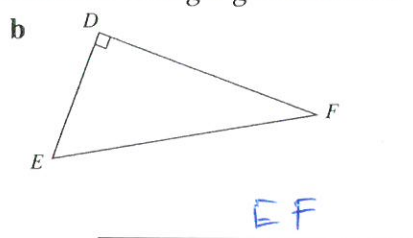
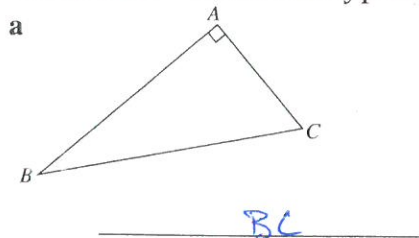
Right-angled triangles and trigonometry

UNIT 2: The trigonometric ratios

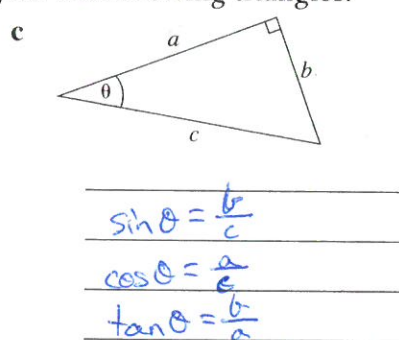
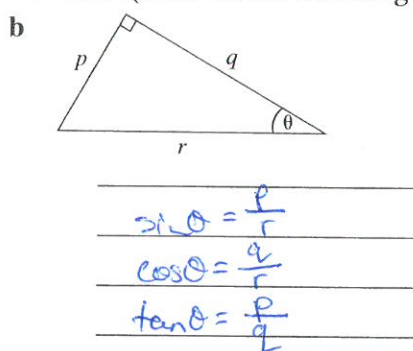
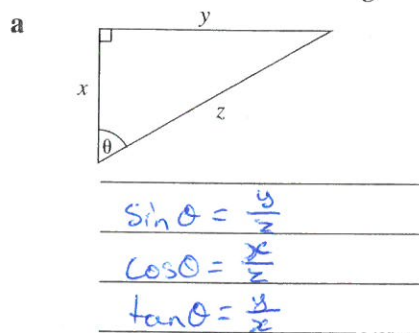
QUESTION 1 In each of the following triangles, state whether a , b and c are the opposite side, adjacent side or hypotenuse with reference to the angle marked.



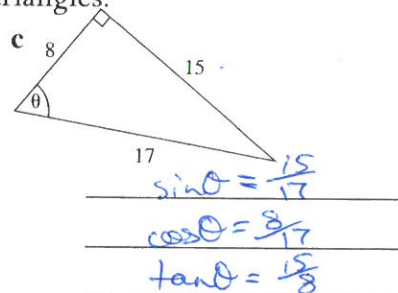
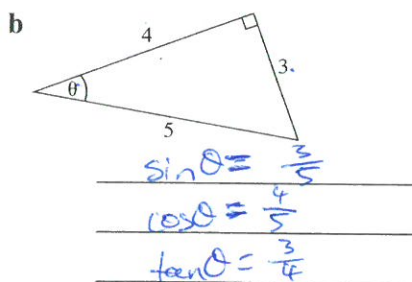
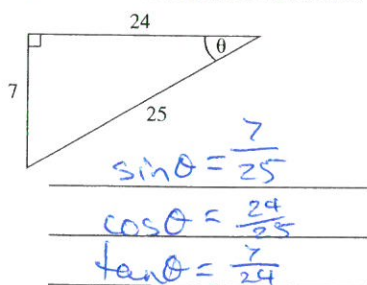
QUESTION 2 Name the hypotenuse in each triangle given below.



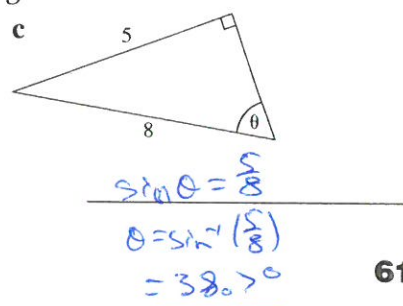
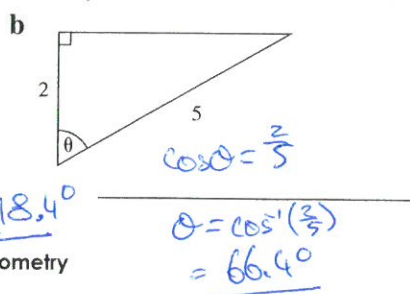
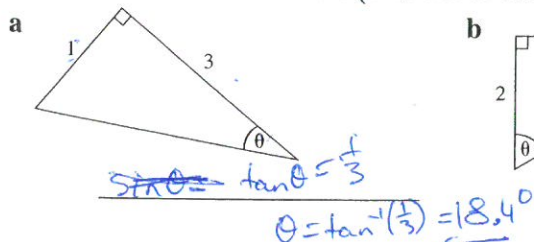
QUESTION 3 Write the trigonometric ratios (sine, cosine and tangent) for the following triangles.



QUESTION 4 Find the fraction $\sin \theta$, $\cos \theta$ and $\tan \theta$ in the following triangles.



QUESTION 5 Which ratio (sin, cos or tan) could be used to find the angle θ .

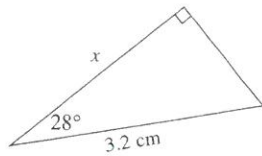


Right-angled triangles and trigonometry

UNIT 4: Finding a side

QUESTION 1 Find the length of the unknown side correct to one decimal place.

a

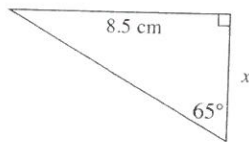


$$\cos 28 = \frac{x}{3.2}$$

$$x = 3.2 \times \cos 28$$

$$x = \cancel{2.81} 2.8 \text{ cm}$$

b

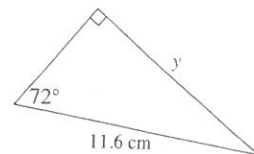


$$\tan 65 = \frac{8.5}{x}$$

$$x = \frac{8.5}{\tan 65}$$

$$x = \cancel{3} 4.0 \text{ cm}$$

c

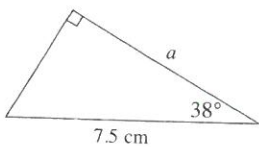


$$\sin 72 = \frac{y}{11.6}$$

$$y = 11.6 \times \sin 72$$

$$y = 11.0 \text{ cm}$$

d

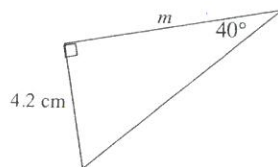


$$\cos 38 = \frac{a}{7.5}$$

$$a = 7.5 \times \cos 38$$

$$a = 5.9 \text{ cm}$$

e

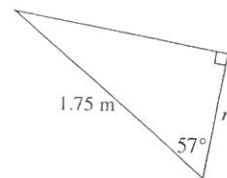


$$\tan 40 = \frac{4.2}{m}$$

$$m = \frac{4.2}{\tan 40}$$

$$m = 5.0 \text{ cm}$$

f



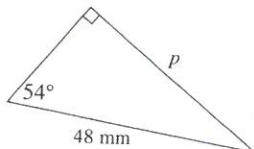
$$\cos 57 = \frac{n}{1.75}$$

$$n = 1.75 \times \cos 57$$

$$n = \cancel{1.0} 1.0 \text{ m}$$

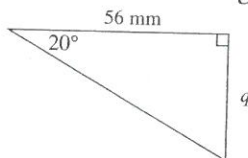
QUESTION 2 Find the value of the pronumeral in each triangle correct to two decimal places.

a



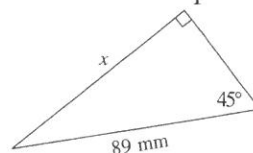
$$p = 38.83$$

b



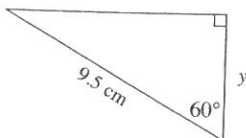
$$q = 20.38$$

c



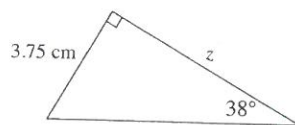
$$x = 62.93$$

d



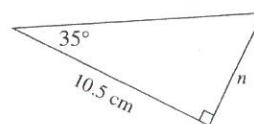
$$y = 4.75$$

e



$$z = 4.80$$

f

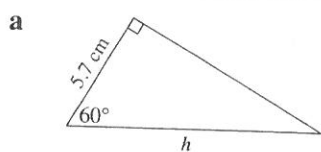


$$n = 7.35$$

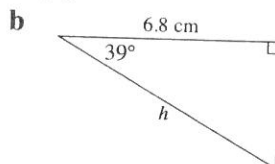
Right-angled triangles and trigonometry

UNIT 5: Finding the hypotenuse

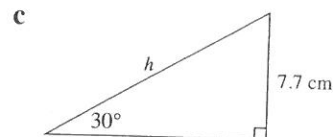
QUESTION 1 Find the length of the hypotenuse correct to two decimal places.



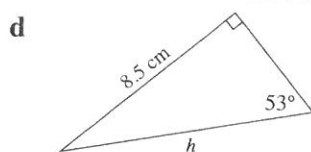
11.40



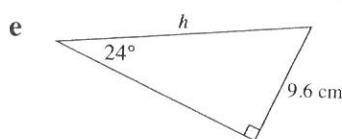
8.75



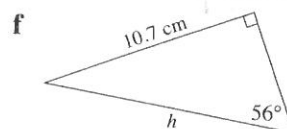
15.40



10.64

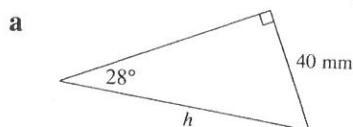


23.60

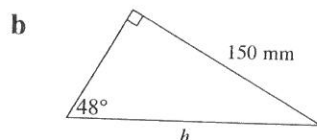


12.91

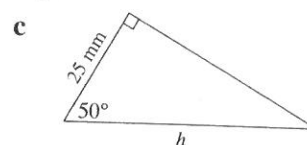
QUESTION 2 Find the length of the hypotenuse correct to one decimal place.



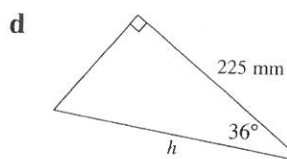
85.2



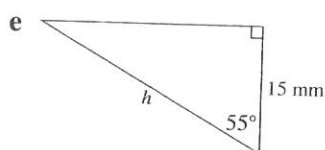
201.8



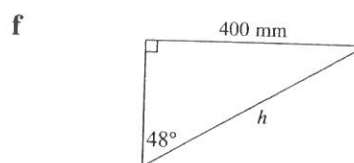
38.9



278.1

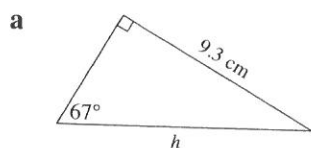


26.2

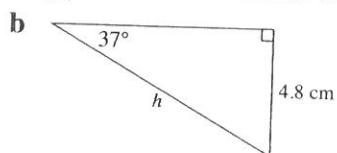


538.3

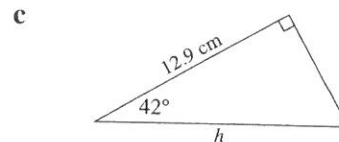
QUESTION 3 Find the length of the hypotenuse correct to two decimal places.



10.10



7.98

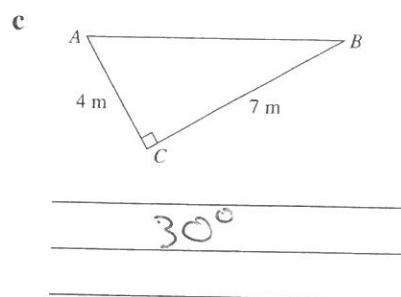
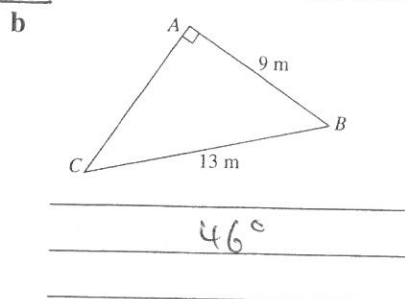
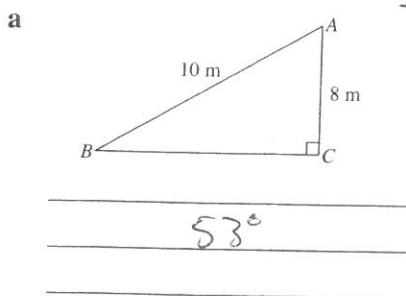


17.36

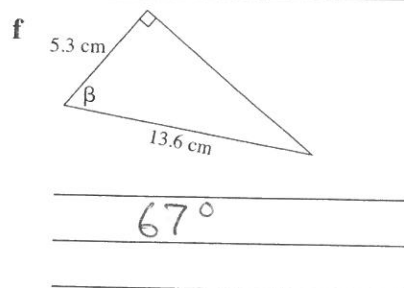
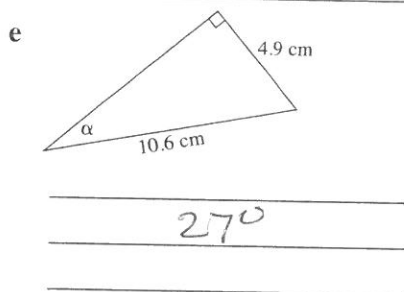
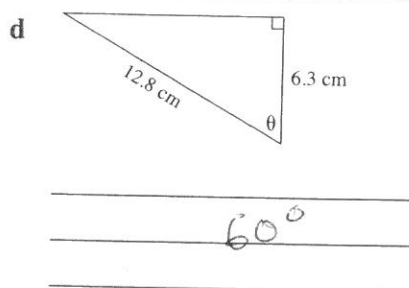
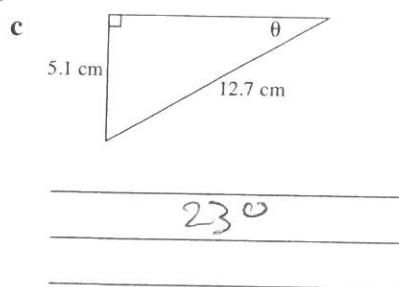
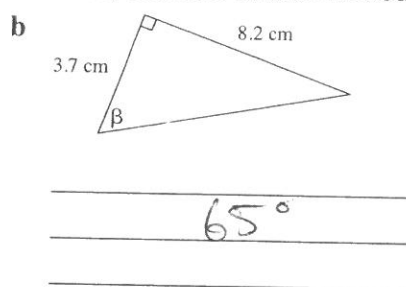
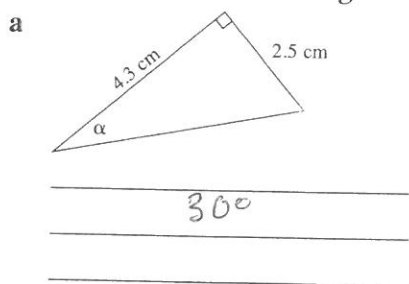
Right-angled triangles and trigonometry

UNIT 6: Finding an unknown angle

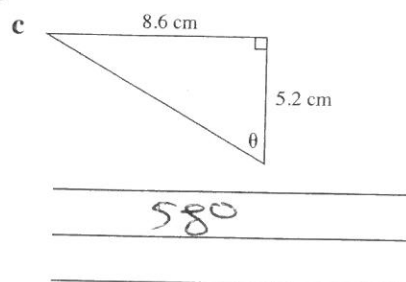
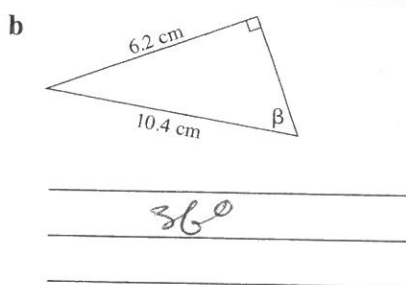
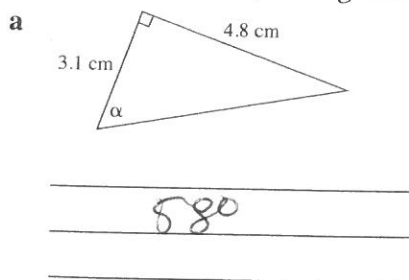
QUESTION 1 Find the size of angle B. Give the answer to the nearest degree.



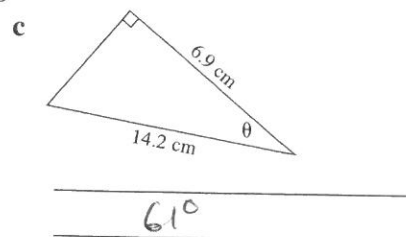
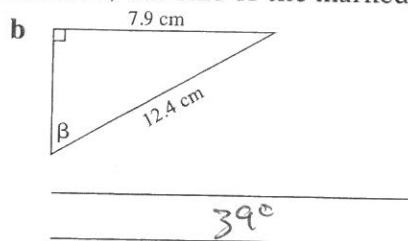
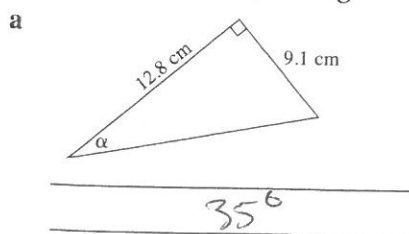
QUESTION 2 Find, in degrees and minutes, the size of the marked angle.



QUESTION 3 Find, in degrees and minutes, the size of the marked angle.



QUESTION 4 Find, in degrees and minutes, the size of the marked angle.



Right-angled triangles and trigonometry

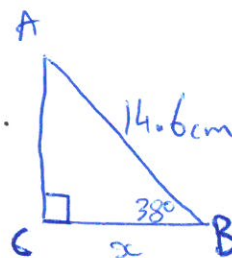
UNIT 7: Mixed problems

- QUESTION 1 In $\triangle ABC$, $\angle C = 90^\circ$, $\angle B = 38^\circ$ and $AB = 14.6$ cm. Find BC correct to one decimal place.

$$\cos 38 = \frac{x}{14.6}$$

$$x = 14.6 \times \cos 38$$

$$x = 11.4 \text{ cm}$$



- QUESTION 2 A ladder leans against a vertical wall with its foot 1.5 metres from the wall making an angle of $45^\circ 36'$ with the ground. How long is the ladder? Give your answer to the nearest centimetre.

$$2.11 \text{ m}$$

- QUESTION 3 A tree 18 metres tall casts a shadow 19.5 metres long. What angle do the rays of the Sun make with the ground?

$$42^\circ$$

- QUESTION 4 The height of a ramp is 4.2 m. Given that the ramp is inclined at 30° to the ground, find the length of the ramp to the nearest centimetre.

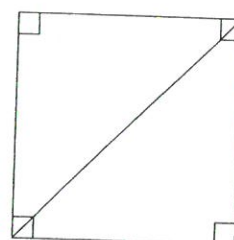
$$8.40 \text{ m}$$

- QUESTION 5 A tree casts a shadow 20.7 m long. If the Sun's rays meet the ground at $29^\circ 36'$, what is the height of the tree to the nearest metre?

$$12 \text{ m}$$

- QUESTION 6 The diagonal of a square is 24.6 cm long. Find the length of one side to the nearest millimetre.

$$17.4 \text{ cm}$$



Further algebra

UNIT 9: Quadratic equations in factorised form

QUESTION 1 Solve the following quadratic equations that are already expressed in factorised form.

a $(x-1)(x-2)=0$

$x=1 \quad x=2$

b $(x-2)(x+3)=0$

$x=2 \quad x=-3$

c $(x-1)(x-3)=0$

$x=1 \quad x=3$

d $x(x+5)=0$

$x=0 \quad x=-5$

e $2x(x-4)=0$

$x=0 \quad x=4$

f $(x-3)(x-7)=0$

$x=3 \quad x=7$

g $(x-3)(x-5)=0$

$x=3 \quad x=5$

h $(x+1)(x-3)=0$

$x=-1 \quad x=3$

i $(x+2)(x-4)=0$

$x=-2 \quad x=4$

j $(x+3)(x-3)=0$

$x=-3 \quad x=3$

k $(x-2)(x+2)=0$

$x=2 \quad x=-2$

l $(x-5)(x+5)=0$

$x=5 \quad x=-5$

m $(x+1)(x-6)=0$

$x=-1 \quad x=6$

n $(x+2)(x+3)=0$

$x=-2 \quad x=-3$

o $x(x+8)=0$

$x=0 \quad x=-8$

QUESTION 2 Solve the following quadratic equations.

a $x(2x-1)=0$

$x=0 \quad 2x-1=0$
 $x=\frac{1}{2}$

b $(x+6)(2x-1)=0$

$x=-6 \quad 2x-1=0$
 $x=\frac{1}{2}$

c $(3x-2)(x+1)=0$

$3x-2=0 \quad x+1=0$
 $x=\frac{2}{3} \quad x=-1$

d $(x-2)(3x-1)=0$

$x=2 \quad x=\frac{1}{3}$

e $5x(2x-1)=0$

$x=0 \quad x=\frac{1}{2}$

f $3x(x-2)=0$

$x=0 \quad x=2$

g $(x+3)(3x-1)=0$

$x=-3 \quad x=\frac{1}{3}$

h $4x(2x-5)=0$

$x=0 \quad x=\frac{5}{2}$

i $-2x(x-1)=0$

$x=0 \quad x=1$

j $(3x+1)x=0$

$x=-\frac{1}{3} \quad x=0$

k $(x-3)2=0$

$x=3$

l $3x(x-3)=0$

$x=0 \quad x=3$

QUESTION 3 Solve the following equations.

a $(x-4)(x-5)=0$

$x=4 \quad x=5$

b $(x-8)(x+8)=0$

$x=8 \quad x=-8$

c $x(x-3)=0$

$x=0 \quad x=3$

d $2x(x-2)=0$

$x=0 \quad x=2$

e $(x-7)(x-9)=0$

$x=7 \quad x=9$

f $(x+1)(x-5)=0$

$x=-1 \quad x=5$

g $(2x-1)(x+4)=0$

$x=\frac{1}{2} \quad x=-4$

h $(2x+3)(2x-3)=0$

$x=-\frac{3}{2} \quad x=\frac{3}{2}$

i $(4x+5)(5x-4)=0$

$x=-\frac{5}{4} \quad x=\frac{4}{5}$

Further algebra

UNIT 11: Solving quadratic equations by factorising

QUESTION 1 Solve the following quadratic equations by factorising.

a $x^2 + 5x + 6 = 0$

$$(x+3)(x+2) = 0$$

$$x = -3 \quad x = -2$$

b $x^2 - 2x - 35 = 0$

$$(x+5)(x-7) = 0$$

$$x = -5 \quad x = 7$$

c $x^2 - 5x - 6 = 0$

$$(x+1)(x-6) = 0$$

$$x = -1 \quad x = 6$$

d $x^2 + 7x + 12 = 0$

$$(x+3)(x+4) = 0$$

$$x = -3 \quad x = -4$$

e $x^2 - 5x + 6 = 0$

$$(x-2)(x-3) = 0$$

$$x = 2 \quad x = 3$$

f $x^2 + 2x - 48 = 0$

$$(x-6)(x+8) = 0$$

$$x = 6 \quad x = -8$$

g $x^2 - 8x + 16 = 0$

$$(x-4)(x-4) = 0$$

$$x = 4$$

h $x^2 + 2x - 15 = 0$

$$x = 3 \quad x = -4$$

i $x^2 + 9x + 20 = 0$

$$x = -4 \quad x = -5$$

j $x^2 - 8x + 15 = 0$

$$x = 3 \quad x = 5$$

k $x^2 + 4x - 12 = 0$

$$x = 2 \quad x = -6$$

l $x^2 - 3x - 10 = 0$

$$x = 5 \text{ or } -2$$

m $x^2 + 11x + 30 = 0$

$$x = -5 \text{ or } -6$$

n $x^2 - 9x + 14 = 0$

$$x = -2 \text{ or } -7$$

o $x^2 + 3x - 28 = 0$

$$x = 11 \text{ or } -9$$

p $x^2 - 2x - 99 = 0$

$$x = 11 \text{ or } -9$$

q $x^2 + 6x + 8 = 0$

$$x = -2 \text{ or } -4$$

r $x^2 + 6x - 7 = 0$

$$x = 1 \text{ or } -7$$

s $x^2 - 6x + 5 = 0$

$$x = 1 \text{ or } 5$$

t $x^2 + 8x + 16 = 0$

$$x = -4$$

u $x^2 - 4x - 60 = 0$

$$x = 10 \text{ or } -6$$

QUESTION 2 Factorise and solve the following quadratic equations.

a $x^2 = 3x + 18$

$$x = 6 \text{ or } -3$$

b $x^2 + 40 = 13x$

$$x = 5 \text{ or } 8$$

c $x^2 + 5x = 36$

$$x = -9 \text{ or } 4$$

d $x^2 = 15x - 54$

$$x = 6 \text{ or } 9$$

e $x^2 - 2x = 24$

$$x = 6 \text{ or } -4$$

f $x^2 = 24 - 5x$

$$x = 3 \text{ or } -8$$

Linear and non-linear relationships

UNIT 4: Parallel and perpendicular lines (1)

QUESTION 1 Complete.

- a Lines that are parallel have gradients that are the same.
- b Lines that are perpendicular have gradients that are negative reciprocal.

QUESTION 2 Determine whether the two given lines are parallel or perpendicular or neither.

- | | | | |
|---|--|--|--|
| a $y = 2x + 3$
$y = 2x - 1$
<u>para</u> | b $y = \frac{1}{2}x + 5$
$y = 2x - 3$
<u>neither</u> | c $y = \frac{3}{2}x + 1$
$y = -\frac{2}{3}x - 1$
<u>perp</u> | d $y = \frac{x}{4} + \frac{2}{3}$
$y = -4x + 2$
<u>perp</u> |
| e $y = 6 - x$
$y = -x + 3$
<u>para</u> | f $y = 1 - 2x$
$y = \frac{x}{2} + 5$
<u>perp</u> | g $y = -\frac{3}{4}x - 2$
$y = -\frac{4}{3}x + 2$
<u>neither</u> | h $y = \frac{5x}{6} - 7$
$y = -\frac{6}{5}x + 1$
<u>perp</u> |

QUESTION 3 Write down the equation of the line that passes through the point (0, 2) and which is parallel to the given line.

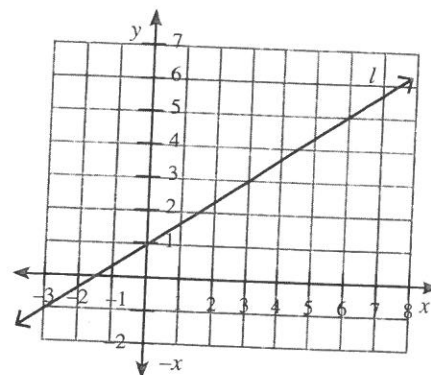
- | | |
|--|--|
| a $y = 3x + 1$
<u>$y = 3x + 2$</u> | b $y = -2x - 3$
<u>$y = -2x + 2$</u> |
| c $y = \frac{1}{2}x + \frac{1}{4}$
<u>$y = \frac{1}{2}x + 2$</u> | d $y = -\frac{5x}{3} - 4$
<u>$y = -\frac{5x}{3} + 2$</u> |

QUESTION 4 Write down the equation of the line that passes through the origin and which is perpendicular to the given line.

- | | |
|---|---|
| a $y = 4x + 3$
<u>$y = -\frac{1}{4}x$</u> | b $y = -\frac{x}{3} + 7$
<u>$y = 3x$</u> |
| c $y = 9 - 2x$
<u>$y = \frac{1}{2}x$</u> | d $y = \frac{3}{2}x + 4$
<u>$y = -\frac{2}{3}x$</u> |

QUESTION 5 The diagram shows the graph of line l .

- a What is the gradient of l ? $\frac{2}{3}$
- b What is the gradient of any line parallel to l ? $\frac{2}{3}$
- c What is the gradient of any line perpendicular to l ? $-\frac{3}{2}$
- d Line m has equation $y = -\frac{3}{2}x - 12$. It intersects line l at P . If l meets the y -axis at Q and m meets the y -axis at R . What is the size of $\angle QPR$? 90°



Linear and non-linear relationships

UNIT 6: Quadratic graphs

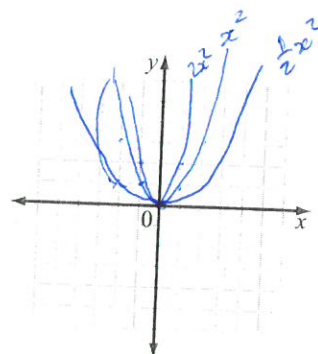
QUESTION 1 Complete the table of values and then, on the same number plane, draw the graphs of the following.

a $y = x^2$

b $y = 2x^2$

c $y = \frac{1}{2}x^2$

x	-3	-2	-1	0	1	2	3
$y = x^2$	9	4	1	0	1	4	9
$y = 2x^2$	18	8	2	0	2	8	18
$y = \frac{1}{2}x^2$	$4\frac{1}{2}$	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2	$4\frac{1}{2}$



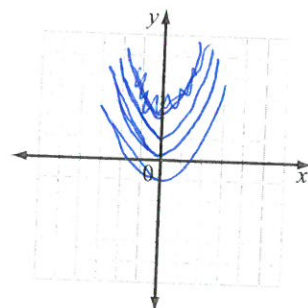
QUESTION 2 Complete the table of values and then, on the same number plane, draw the graphs of the following.

a $y = x^2$

b $y = x^2 + 1$

c $y = x^2 - 1$

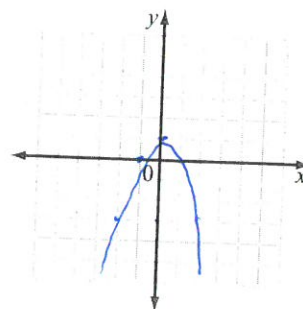
x	-3	-2	-1	0	1	2	3
$y = x^2$	9	4	1	0	1	4	9
$y = x^2 + 1$	10	5	2	1	2	5	10
$y = x^2 - 1$	8	3	0	-1	0	3	8



QUESTION 3 Complete the table of values for $y = 1 - x^2$ and sketch its graph.

x	-3	-2	-1	0	1	2	3
$1 - x^2$	-8	-3	0	1	0	-3	-8

- a What is the equation of its axis of symmetry? $x=0$
- b What are the coordinates of its vertex? $(0,1)$
- c What is the maximum value for $y = 1 - x^2$? $y=1$
- d Find the x -intercepts. -1 and 1
- e Find the y -intercept. $y=1$



QUESTION 4 Sketch the graphs of the following.

a $y = x^2$

b $y = x^2 + 2$

c $y = x^2 - 2$

d Explain how the graphs of $y = x^2 + 2$ and $y = x^2 - 2$ can be drawn using $y = x^2$.

sketching $y = x^2 + 2$ move parabola 2 units up on y-axis and for $y = x^2 - 2$ go down 2

